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Program: Software Engineering

Operating Systems

Laboratory Work No. 1

The OS as a Resource Manager

Track A: The Real OS, from the Outside (Linux)

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1. Purpose of the Work

The purpose of this laboratory work is to observe how a real operating system manages files, processes, memory, devices, and storage. The experiments were performed in Ubuntu under WSL2 on the student's own computer.

2. System Information

```
Linux DESKTOP-10DB5P6 5.15.167.4-microsoft-standard-WSL2 #1 SMP Tue Nov
    5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
17:26:07 up 2 min, 1 user, load average: 0.01, 0.00, 0.00
```

Figure 1.1 – Actual uname and uptime output from the laboratory environment

3. Part 1. Files and Directories

A test file was created and its permissions were changed with `chmod`.

```
total 8
-rw-r--r-- 1 sofia sofia 24 Sep 28 17:28 note.txt
-rw-r--r-- 1 sofia sofia 24 Sep 28 17:28 renamed.txt
```

Figure 2.1 – Actual listing of files created during the experiment

3.1. Observe answers

1. The created files are owned by user `sofia` and group `sofia`.
2. After `chmod 600 note.txt`, the mode became `-rw-------`: the owner can read and write, while group and other users have no permissions.
3. Two directories directly under `/` are `/etc`, which stores system-wide configuration, and `/home`, which stores user home directories.

4. Part 2. Processes

A `sleep` process was started and inspected through `/proc`.

```
Name:   sleep
State:  S (sleeping)
Pid:    665
PPid:   287
VmPeak:     3128 kB
VmSize:     3128 kB
```

Figure 3.1 – Actual excerpt from /proc/665/status

4.1. Observe answers

1. Process 1 had PID 1. In this WSL2 environment it was `/sbin/init`, the init process.
2. Approximately 29 processes were running on the idle system during the observation.
3. The `sleep` process was in state `S (sleeping)`, meaning it was waiting and was not actively consuming CPU time.

5. Part 3. Memory

```
               total        used        free      shared  buff/cache
   available
Mem:            13Gi       711Mi        12Gi       3.3Mi       524Mi
        12Gi
Swap:          4.0Gi          0B       4.0Gi
```

Figure 4.1 – Actual memory and swap information reported by free

5.1. Observe answers

1. The system exposed about 13 GiB of RAM, with about 12 GiB free at the time of the experiment.
2. Swap is disk-backed space the operating system may use when physical memory is under pressure. The environment had 4.0 GiB configured and it was unused at that moment.
3. The `sleep` process used 1080 kB of resident memory (`VmRSS`), which is small because the program performs almost no work.

6. Part 4. Devices and Storage

```
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 388.4M  1 disk
sdb    8:16   0     4G  0 disk [SWAP]
sdc    8:32   0     1T  0 disk /mnt/wslg/distro
                               /
```

Figure 5.1 – Actual block-device layout reported by lsblk

6.1. Observe answers

1. The root filesystem `/` is mounted on `/dev/sdc`.
2. `/dev/console` is an example of a device exposed through the filesystem; it represents the system console.
3. The Unix idea that “everything is a file” means that many kernel-managed resources, including devices, are accessed through file-like interfaces.

7. Conclusion

The experiments showed the operating system managing files, processes, memory, devices, and storage. Commands such as `ls`, `ps`, `free`, `lsblk`, and the `/proc` filesystem exposed the state of these resources and demonstrated how Linux presents them through consistent interfaces.